

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365750893	delta=+0.0141533
ratio=1.0060186		
CoG x [m]	-0.002024708562 -> -0.002196564731	delta=-0.000171856
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008635520624	delta=+0.000894079
ratio=-		
CoG z [m]	0.009509749599 -> 0.01053734438	delta=+0.00102759
ratio=-		
rotor effectiveness 1	1 -> 0.9669377633	delta=-0.0330622 ratio=0.96693776
rotor effectiveness 2	1 -> 0.9720486674	delta=-0.0279513 ratio=0.97204867
rotor effectiveness 3	1 -> 0.9576034214	delta=-0.0423966 ratio=0.95760342
rotor effectiveness 4	1 -> 0.9560470593	delta=-0.0439529 ratio=0.95604706
rotor command lag [s]	0.004997730255 -> 0.2	delta=+0.195002 ratio=-

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501027634, -2.420837172e-05, 2.303452102e-05]
[-2.420837172e-05, 0.06523926317, 6.493518769e-06]
[ 2.303452102e-05, 6.493518769e-06, 0.1289936903]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06500773633	delta=+5.50912e-05
ratio=1.0008482		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06524179423	delta=+0.000241727
ratio=1.0037189		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289936992	delta=+1.5829e-06
ratio=1.0000123		

Optimization

joint dynamics loss: 1.09354357258
Gaussian-prior cost: 0.331330655476
selected strict-ZOH rotor lag [s]: 0.2

Bag failure_1_single

reconstruction position RMSE [m]: 0.000497212640177
reconstruction position component RMSE [m]: [0.000302 0.000297 0.000261]
reconstruction orientation RMSE [deg]: 0.528821719301
gyro consistency RMSE xyz [rad/s]: [0.060795 0.087547 0.042573]
specific-force consistency RMSE xyz [m/s^2]: [0.937288 0.202709 0.378337]
inferred external force RMS xyz [N]: [0.978972 0.534736 3.019088]
inferred external torque RMS xyz [N m]: [0.121858 0.283449 0.209839]
bag dynamics loss: 1.09354357258
gimbal command lag [s] 0.004998683929 -> 0.03825060843

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message=`gtol` termination condition is satisfied.
reported optimality=2.3125099168908037e-08
initial cost=1.4248742718942367
final cost=1.4248742280551663
cost reduction=4.3839070373863365e-08
coordinate step L2=5.7280483137447645e-05
final gradient inf-norm=2.3125099168908037e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=409.998268653704
max directional finite-difference Jacobian relative error=9.175095543339716e-12